

DOMINIQUE DANG

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Education

Massachusetts Institute of Technology

May 2027

B.Sc. Computer Science & Molecular Biology

Classwork: Biostatistics, Computational Biology, Biological and Engineering Principles Underlying Novel Biotherapeutics, Machine Learning, Algorithms, Linear Algebra, Differential Equations

Research & Work Experience

Pharmacometrician Intern *Eli Lilly*

May – Aug. 2026

- Designed and led a systematic evaluation of Claude AI's ability to independently execute PK/PD and population PK (PopPK) modeling workflows, addressing a key MIDD accessibility gap in drug development
- Benchmarked AI-guided NCA, PK/PD, and PopPK modeling outputs against expert pharmacometric analyses using PK/PD/MIDD principles, quantifying accuracy, reasoning validity, and workflow completeness across 3 case studies

Oncology Data Science Intern *Novartis Biomedical Research Institute*

May – Aug. 2025

- Developed a predictive ML model in Python/R to forecast tumorigenesis outcomes by integrating differential expression and enrichment features, uncovering 200+ biomarkers significantly associated with tumor formation
- Applied gene signature scores to single-cell RNA-seq data from human breast cancer tumors to identify malignant subpopulations with high engraftment potential

Undergraduate Research *Prof. Nir Hacohen - Broad Institute*

Aug. 2025 – Present

- Analyze single-cell RNA-seq data from cutaneous and mucosal melanoma to identify cellular and molecular mechanisms underlying differential immunotherapy responses
- Develop gene co-occurrence networks from pan-cancer CRISPR screens to cluster immunosuppressive genes and predict functional relevance of hits from targeted CRISPR screens

Publications

J. Clarin, **D. Dang**, L. Santos, and R. Amini. Mechanical characterization of porcine tricuspid valve anterior leaflets over time: Applications to ex vivo studies. *ASME Open Journal of Engineering*, 2:021032, 05 2023. ISSN 2770-3495. doi:[10.1115/1.4062477](https://doi.org/10.1115/1.4062477). URL <https://doi.org/10.1115/1.4062477>.

Leadership & Extracurriculars

HackMIT *Logistics Director*

Feb. 2024 – Present

- Led a 15-member subteam to organize HackMIT, one of the largest U.S. collegiate hackathons, enhancing operational efficiency and streamlining the review process for 3,000+ applications

dynaMIT *Co-Director*

Sept. 2023 – Present

- Led operations and curriculum planning for dynaMIT, a STEM outreach program serving 80+ middle school students and coordinating 40+ mentors

Technical Skills

Programming Python (Pandas, scikit-learn, NumPy, PyTorch), R (tidyverse, ggplot2, Seurat Markdown), MATLAB, Bash, Git, LaTeX

Modeling PK/PD Modeling, Population PK Modeling, Noncompartmental Analysis, MIDD, Machine Learning, Statistical Modeling

Bioinformatics Single-Cell RNA-seq Analysis, DESeq2, SAMtools, IGV, FASTQ/FASTA

Awards

1st place, Innovation in Research: Data Science Received at Novartis Biomedical Research Institute. **2025**

Genetic Medicine Scholar

2025

Selected as 1 of 12 MIT sophomores for Eli Lilly's competitive research fellowship, focusing on disease pathology mapping and early-stage drug development.

Top Beginner Hack Received at HackMIT, one of the largest collegiate hackathons in the US.

2023